andrew@ee.iit.ac.in Q1: 25% Q2: 25% F: 50%

Probability and Statistics (DeGroot and Scherwish)

(Grimmet and Stirzaker)

Sets:

Finite Set: S: set |S|: no. of elements or size

$|S| < \infty$

Countable set: $S = \{a_1, a_2, a_3, \ldots\}$

examples: set of integers, rational numbers

set of all real numbers from 0 to 1 is not countable (uncountable sets)

(proved by contradiction)

Function: maps domain to codomain

$S = \{1,2,3,4,5,6\}$

2^S (power-set) = set of all subsets of S .

Experiment: phenomenon / event we wish to study

Sample Space: set of all outcomes of the experiment

toss coin twice: Sample space = $\{HH, HT, TH, TT\}$

Event: subset of a sample space

$|S| = n \Rightarrow$ No of possible events $= 2^n$

Conditions:

1. Sample space should be an event
2. If A is an event $\Rightarrow A^c$ is an event
3. If $A_1, A_2, \ldots$ are countable collection of events, then $A_1 \cup A_2 \cup \ldots$ is also an event

Event Occured: Outcome belongs to the event

Probability: function from collection of events to $[0,1]$

Probability space: Sample Space; Collection of events; probability measure (usually power set)

Axioms for Probability:

1. $P(A) \geq 0$ of any event A
2. $P(S) = 1$
3. $A_1, A_2, \ldots$ infinite sequence of disjoint sets
   $P(A_1 \cup A_2 \ldots) = P(A_1) + P(A_2) + \ldots$

$P(S \cup \phi) = P(S) + P(\phi) \Rightarrow \boxed{P(\phi) = 0}$

$P(A \cup A^c) = P(S) = P(A) + P(A^c)$

$\Rightarrow \boxed{P(A^c) = 1 - P(A)}$

$\boxed{\text{If } B \subseteq A \Rightarrow P(B) \leq P(A)}$



$\boxed{P(A) \geq 0 \; ; \quad P(A) \leq 1}$ (A is a subset of S)

To prove:

$P(A) + P(B) = P(A \cup B) + P(A \cap B)$

$A = (A \cap B) \cup (A \cap B^c) \qquad B = (A \cap B) \cup (A^c \cap B)$

$A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$

$P(A \cup B) = P(A \cap B) + P(A \cap B^c) + P(A^c \cap B)$

$P(A) = P(A \cap B) + P(A \cap B^c) \qquad P(B) = P(A \cap B) + P(A^c \cap B)$

substitute them to prove

Finite Sample Space:

① Coin toss: $S = \{H, T\}$

Events = $\phi, \{H\}, \{T\}, \{H, T\}$

P : 0 p 1-p 1

② Die roll: $S = \{1, 2, 3, 4, 5, 6\}$

Events: $\emptyset$ {1} {2} {3} {4} {5} {6} S

0 P_1 P_2 P_3 P_4 P_5 P_6 1

$\sum_{i=1}^{6} P_i = 1$, $0 \le P_i \le 1$

Other events can be written as disjoint union of these 6 singleton events.

Usual assumptions are that coins/die are fair.

③ Draw a card from shuffled deck.

52 cards

4 suits: heart, spades, diamonds, clubs

A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K

S = {A heart, 2 heart, ... K clubs}

P = 1/52 1/52 1/52

each singleton element has 1/52 probability.

④ Marbles in urns:

5 black balls, 8 blue balls

S = {black$_1$, black$_2$, ... blue$_8$}

P = 1/13 1/13 ..... 1/13

(or) S = {black, blue} ⇒ E = $\emptyset$ black blue S

P = 5/13 8/13 P = 0 5/13 8/13 1

⑤ Throw a die twice (it's fair)

$S = \{(1,1), (1,2) \ldots\ldots (6,6)\}$

$P(\text{Sum} = 7) \Rightarrow \frac{6}{36}$ $\quad$ $P(\text{two are unequal}) \Rightarrow \frac{30}{36}$

120 boys 30 girls : Commitee of 6B, 4G

$$\Rightarrow \binom{120}{6}\binom{30}{4}$$

Commitee of 10 is made, find probab of 6B+4G

$$\Rightarrow P = \frac{\binom{120}{6}\binom{30}{4}}{\binom{150}{10}}$$

Donut shop; 10 varieties, 12 in a box

ordered: 10^{12}

unordered: no of non-ve integer solutions of

$$a_0 + a_1 + \cdots + a_9 = 12 \Rightarrow {}^{21}C_9$$

Binomial Theorm: $(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}$

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n$$

Multinomial Theorm:

→ Pick 1 subset (size = K) $\Rightarrow {}^nC_K$

→ Pick 1 subset of size K_1 & then another subset of size K_2 $\Rightarrow \frac{n!}{K_1! K_2! (n-K_1-K_2)!}$

52 cards dealt to 4 players (13 each)

$${}^{52}C_{13} \times {}^{39}C_{13} \times {}^{26}C_{13} \times {}^{13}C_{13} = \frac{52!}{13!\,13!\,13!\,13!}$$

spades are divided into 4,3,3,3 : $\dfrac{\frac{39!}{9!10!10!10!}\cdot\frac{13!}{4!3!3!3!}\times {}^4C_3}{\frac{52!}{13!13!13!13!}}$

Counting unions:

4 bins 10 balls each ball put in ^{4} bin at random

to track no of balls in each bin.

sample space: $^{13}C_3$ $\quad$ E → no empty bin

$A_i \to$ ith bin is empty $\quad E = A_1^c \cap A_2^c \cap A_3^c \cap A_4^c = E$

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 4 \cdot 3^{10}/4^{10} - 6 \cdot 2^{10}/4^{10} + 4 \cdot 1^{10}/4^{10}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$P(A_1 \cup A_2 \dots \cup A_n) = \sum_{i=1} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum\sum\sum_{i<j<k} P(A_i \cap A_j \cap A_k)$$

$$\vdots$$

$$(-1)^{n-1} P(A_1 \cap A_2 \dots \cap A_n)$$

Derangement

n-letters, n-addressed envelopes (letter i in i^{th} box)

π: Pick permutation of $[n]$ uniformly at random

letter $i \longrightarrow$ envelope $\pi(i)$

E: no letter goes into correct envelope

A_i: ith letter goes into envelope i

$$E = A_1^c \cap A_2^c \dots \cap A_n^c = (A_1 \cup A_2 \cup \dots \cup A_n)^c$$

$P(A_i) = \frac{(n-1)!}{n!} = 1/n$

$P(A_i \cap A_j) = \frac{1}{(n-1)n}$ $\quad i \neq j$

$\Rightarrow P(E) = 1 - 1/1! + 1/2! - 1/3! \cdots (-1)^n/n!$

$n \to \infty \Rightarrow P(E) = 1/e$

Conditional Probability:

Lottery: Pick a subset of size 5 from $\{1, 2, \ldots 30\}$

$\{a_1, a_2, a_3, a_4, a_5\}$

lottery manager chooses: $\{b_1, b_2, b_3, b_4, b_5\}$

A, B are events; B has occured $(P(B) > 0)$

Conditional probability $P(A|B)$ { probability of A given B

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Response	Imiprimime	Lithium	Combination	Placebo	Total
Relapse	18	33	22	24	77
No-relapse	22	25	16	10	73
Total	40	38	38	34	150

Multiplication Rule:

$$P(A \cap B) = P(A|B)\,P(B) = P(B|A)\cdot P(A)$$

$$P(A_1 \cap A_2 \cap \dots A_n) = P(A_1)\,P(A_2|A_1)\,P(A_3|A_1 \cap A_2)\cdots P(A_n|A_1 \cap \dots A_n)$$

$$P(A_1 \cap A_2 \cdots \cap A_n|B) = P(A_1|B)\cdot P(A_2|A_1 \cap B)\,P(A_3|A_1 \cap A_2 \cap B)\cdots$$

$$\cdots P(A_n|A_1 \cap A_2 \cdots A_{n-1} \cap B)$$

$$P(A|A) = 1$$

Birthday problem (n-people):

$$P(\text{no two have same birthday}) = 1\cdot\left(1-\frac{1}{365}\right)\left(1-\frac{2}{365}\right)\cdots\left(1-\frac{n-1}{365}\right)$$

$A_i \to i^{th}$ person doesn't share b-day with previous people

Partitions:

Experiment: Pick an urn; Pick a marble at random from chosen urn (A: marble is red)

$n_1 R_1 ; m_1 B_1$ $\quad n_2 R_2 ; m_2 B_2$

$$P(A) = P(A|urn1)\,P(urn1) + P(A|urn2)\,P(urn2) = \left(n_1/n_1+m_1 + n_2/n_2+m_2\right)\frac{1}{2}$$



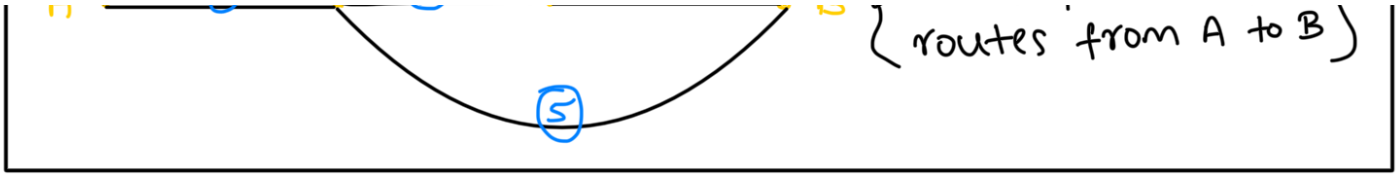
$$A = (A \cap S_1) \sqcup (A \cap S_2) \cdots \sqcup (A \cap S_k)$$

↑ disjoint union

$$(S = S_1 \sqcup S_2 \sqcup \cdots \sqcup S_k)$$



{No of possible open



Monte Hall problem: Gameshow ; 3 doors

1 door → car behind ; 2 doors → Goats

Host opens door 3 + there's a goat behind it.

You can switch / stay. (we switch)

assumption: host knows what's behind each door.

all 6 permutations are equally likely

$G_1 \leftrightarrow G_2 \rightarrow C$

host will always show you a goat:

①	②	③	pick	host	
G_1	G_2	C	①	②	$p = 1/6 \times 1/3$
G_1	G_2	C	②	①	$p = 1/6 \times 1/3$
G_1	G_2	C	③	①	$p = 1/6 \times 1/3 \times 1/2$
G_1	G_2	C	③	②	$p = 1/6 \times 1/3 \times 1/2$
⋮	⋮	⋮			

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we can use symmetry $\Rightarrow \dfrac{1/6 \times 1/3 \times 2}{1/6 \times 1/3 \times 3} = 2/3$

Independence: Two events A and B are said to be independent if $P(A \cap B) = P(A) \cap P(B)$

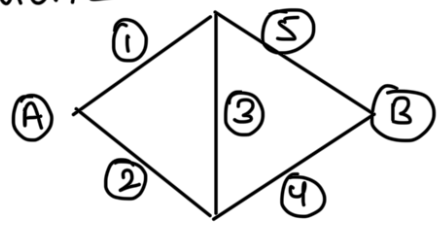
disjoint events → always independent

→ independence of multiple events:

$$P(A_1 \cap A_2 \dots \cap A_n) = P(A_1)\,P(A_2) \dots P(A_n)$$

pairwise independent ← $P(A_i \cap A_j) = P(A_i) \cap P(A_j) \quad (i \neq j)$ — mutually independent

Question 1



E: Roads $R_1, R_2, \dots R_i$ are open (i-roads) and others are shut $P(E) = p^i (1-p)^{5-i}$ F: route from A to B exists

$$P(F) = \sum P(F/E_i) \cdot P(E_i) = \sum P(F \cap E_i)$$

$i=0 \Rightarrow 0$

$i=1 \Rightarrow 0$

$i=2 \Rightarrow 2p^2(1-p)^3$

$i=3 \Rightarrow 8p^3(1-p)^2$

$i=4 \Rightarrow 5p^4(1-p)$

$i=5 \Rightarrow p^5$

Bayes Theory:

$$P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$$

$$\Rightarrow P(B|A) = P(A|B)\frac{P(B)}{P(A)}$$

sphere, surface painted with red & blue

Blue makes up 10% of the surface.

Is there an inscribed cube, all of whose vertices are red.

P(vertex at blue after rotation) = 0.1

$\Rightarrow$ P(all at blue) ≤ 0.8 $(A_1 \cup A_2 \cup A_3 \ldots \cup A_8 \leq 0.8)$

$\Rightarrow$ P(all at red) ≥ 0.2

A, B, C, D : graduates p: any two become friends

A $\rightarrow$ hears rumor Probability (D will hear it)